

# Is a medal about the athlete, or the sport?

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# Problem statement

Across 261,507 Olympic athlete records from 1896 to 2016, win rate ranges from 4.9% in Alpine Skiing to 28.7% in Hockey among sports with more than 5,000 records, a nearly 6x gap. Judging sports on results alone, without accounting for this gap, can mislead decisions on where to develop athletes.

# Aim

Show how much medal odds differ between Olympic sports, and identify which sports have the highest and lowest win rates.



# Objectives

- What is the overall win rate across all athlete records, 1896-2016, and how many sports and events does it cover?
- How do the 18 sports with 5,000+ records rank by win rate, from highest to lowest?
- What is the win rate in Summer vs Winter Games, and how large is the gap?
- How does win rate differ across age, height and weight groups?
- Do sex, season, age, height and weight change which medal (Gold, Silver or Bronze) an athlete wins?

# Dataset overview



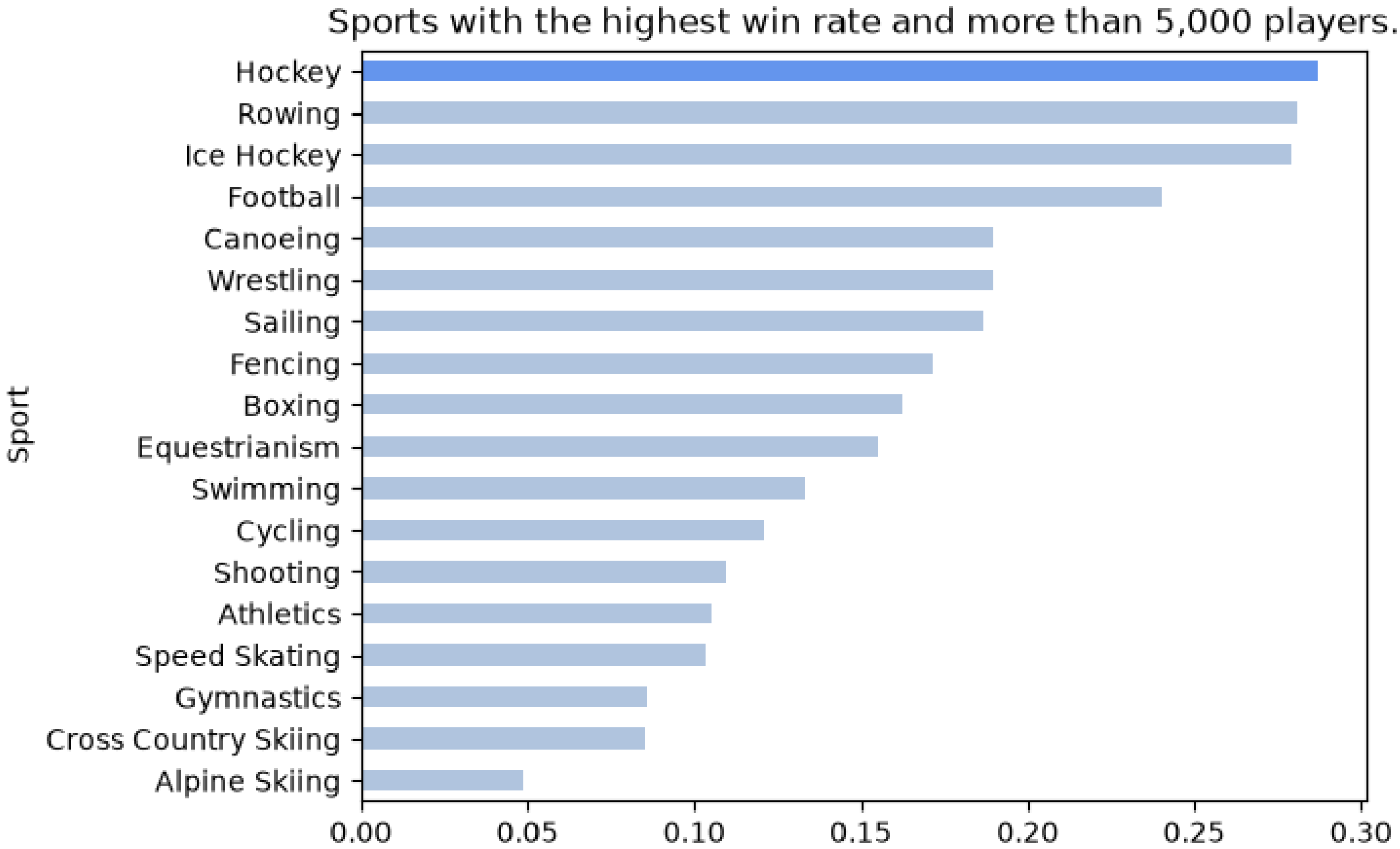
ATHLETE RECORDS :
**261,507**



OVERALL WIN RATE:
**14.9%**

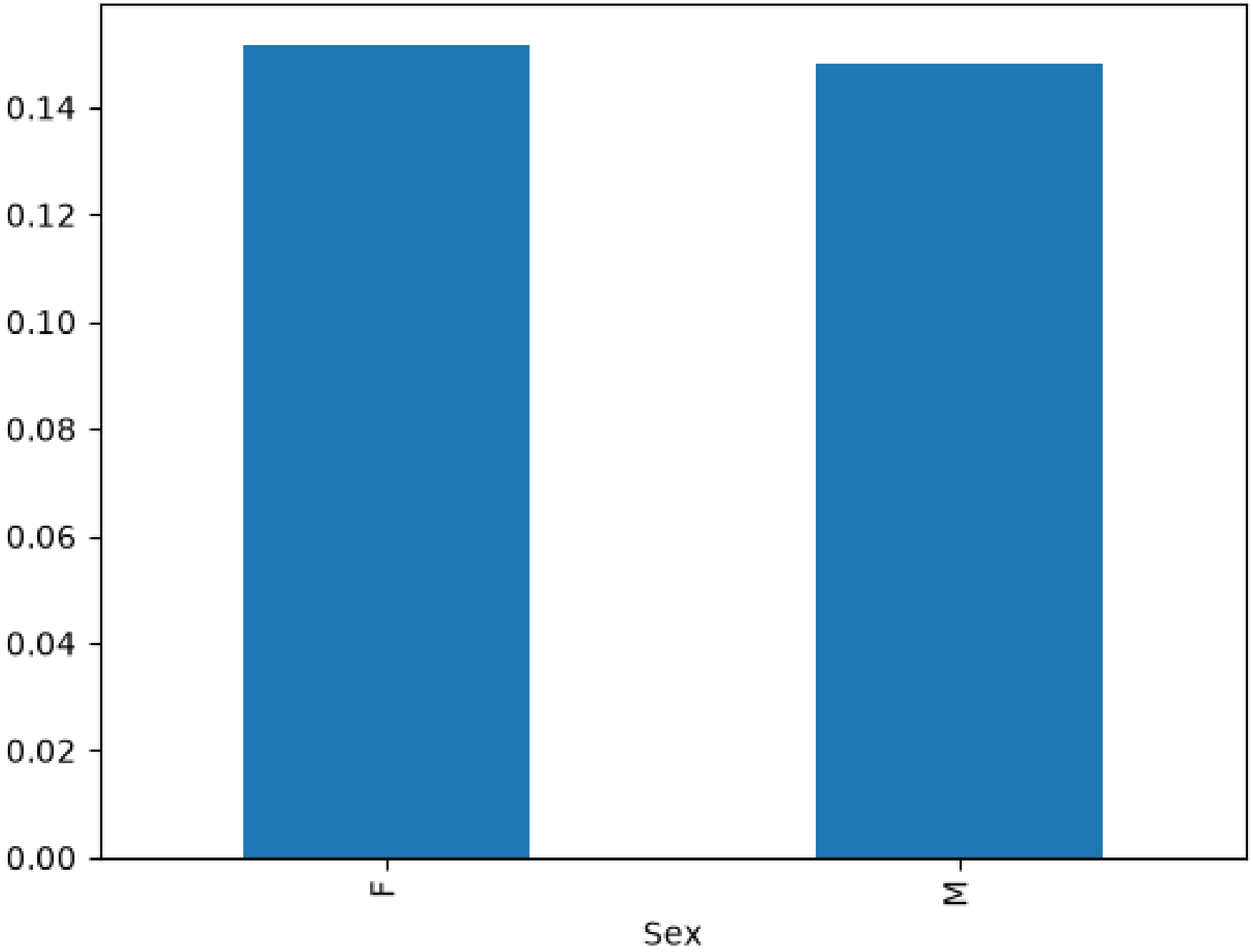


TOTAL SPORTS:
**66**


Sports with the highest win rate and more than 5,000 players.
Sport
Hockey
Rowing
Ice Hockey
Football
Canoeing
Wrestling
Sailing
Fencing
Boxing
Equestrianism
Swimming
Cycling
Shooting
Athletics
Speed Skating
Gymnastics
Cross Country Skiing
Alpine Skiing
0.00
0.05
0.10
0.15
0.20
0.25
0.30

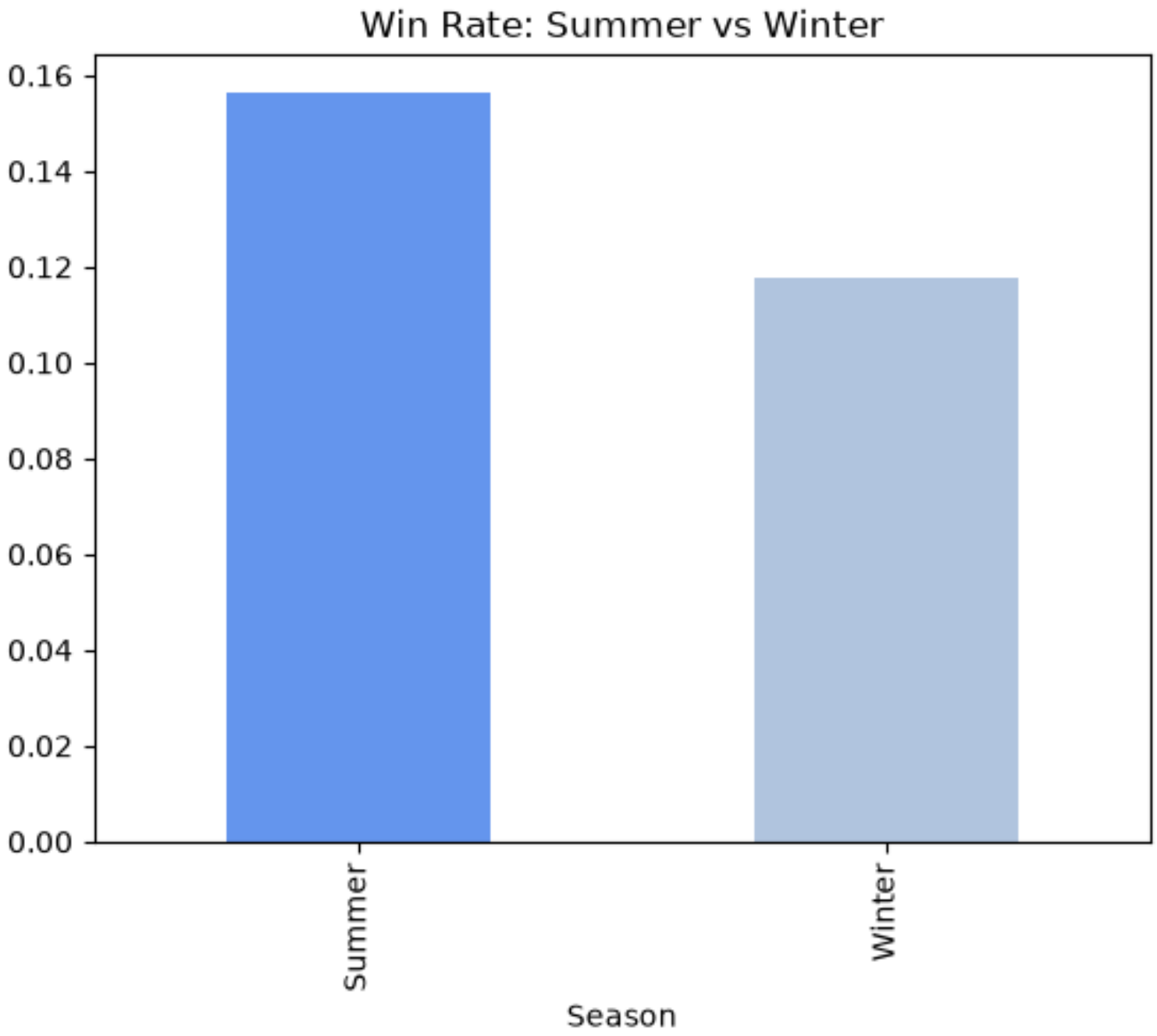
# Findings: sports

- Hockey (28.7%), Rowing and Ice Hockey (both about 28%) and Football (about 24%) lead.
- All four leaders are team sports, where one medal counts for the whole roster.
- Alpine Skiing (4.9%), Cross Country Skiing (about 8.5%) and Gymnastics (about 8.6%) sit at the bottom.
- The spread from top to bottom is nearly 6x.

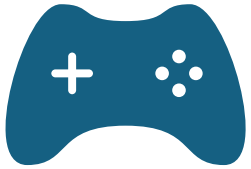

0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
F
M
Sex

# Finding: win rate by sex

Women win medals at **15.1%** and men at **14.8%**, a gap of only **0.3** percentage points. Sex has almost no effect on the chance of winning.


Win Rate: Summer vs Winter
0.16
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
Summer
Winter
Season

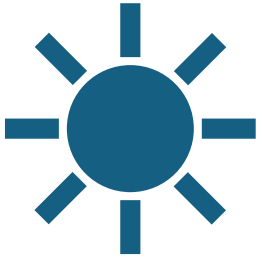
# Findings: season



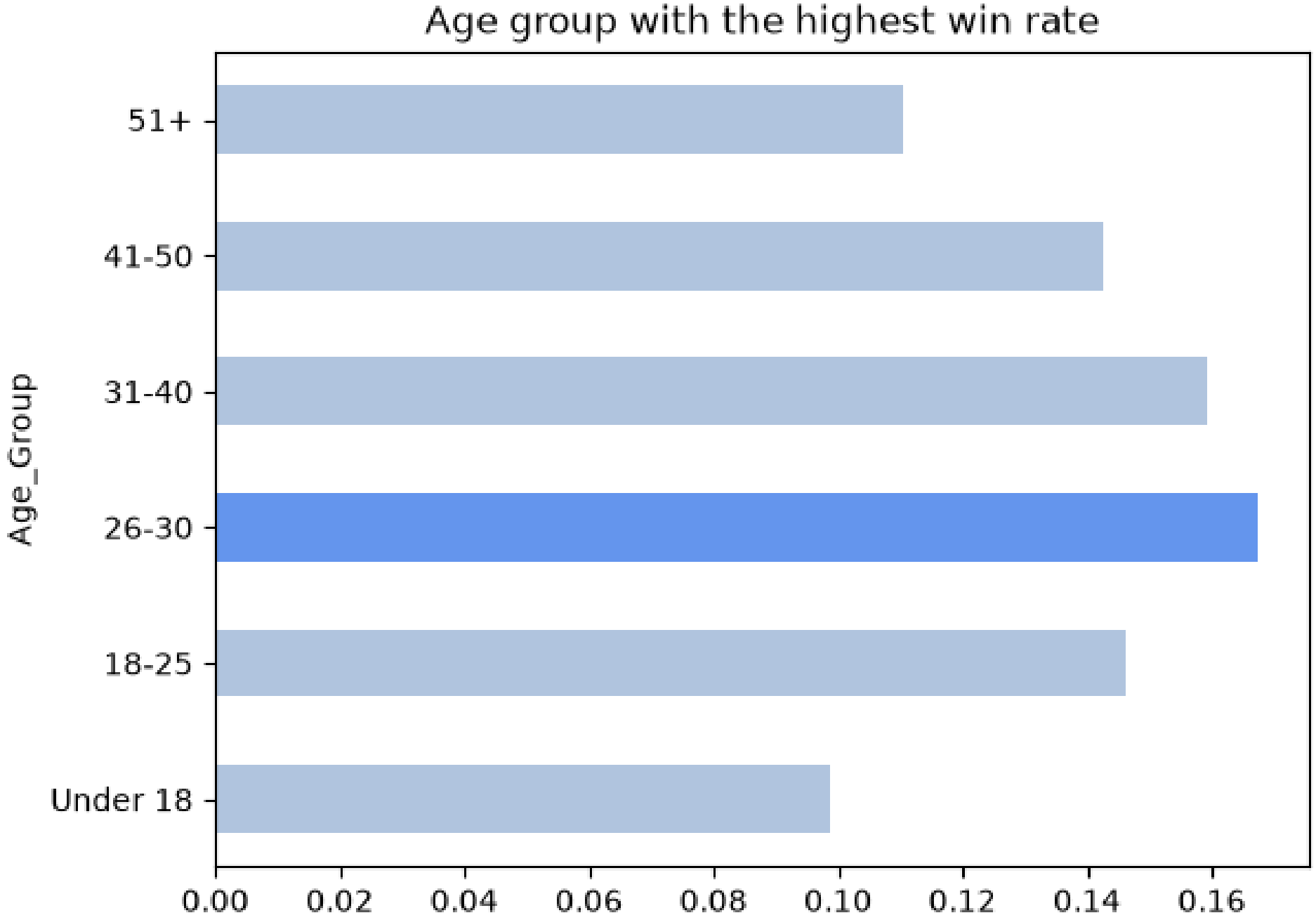
Summer Games win rate is about 15.6%.

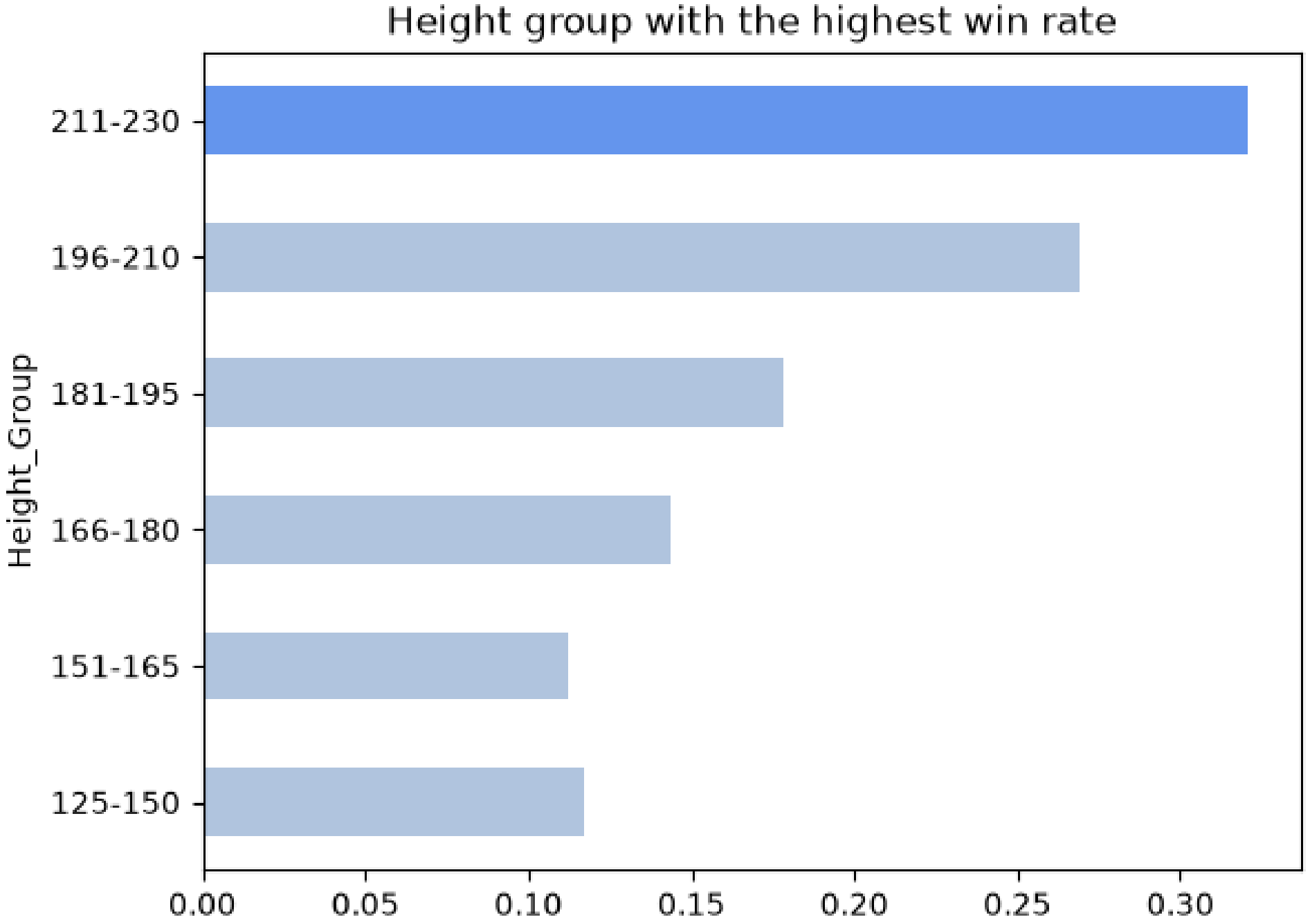


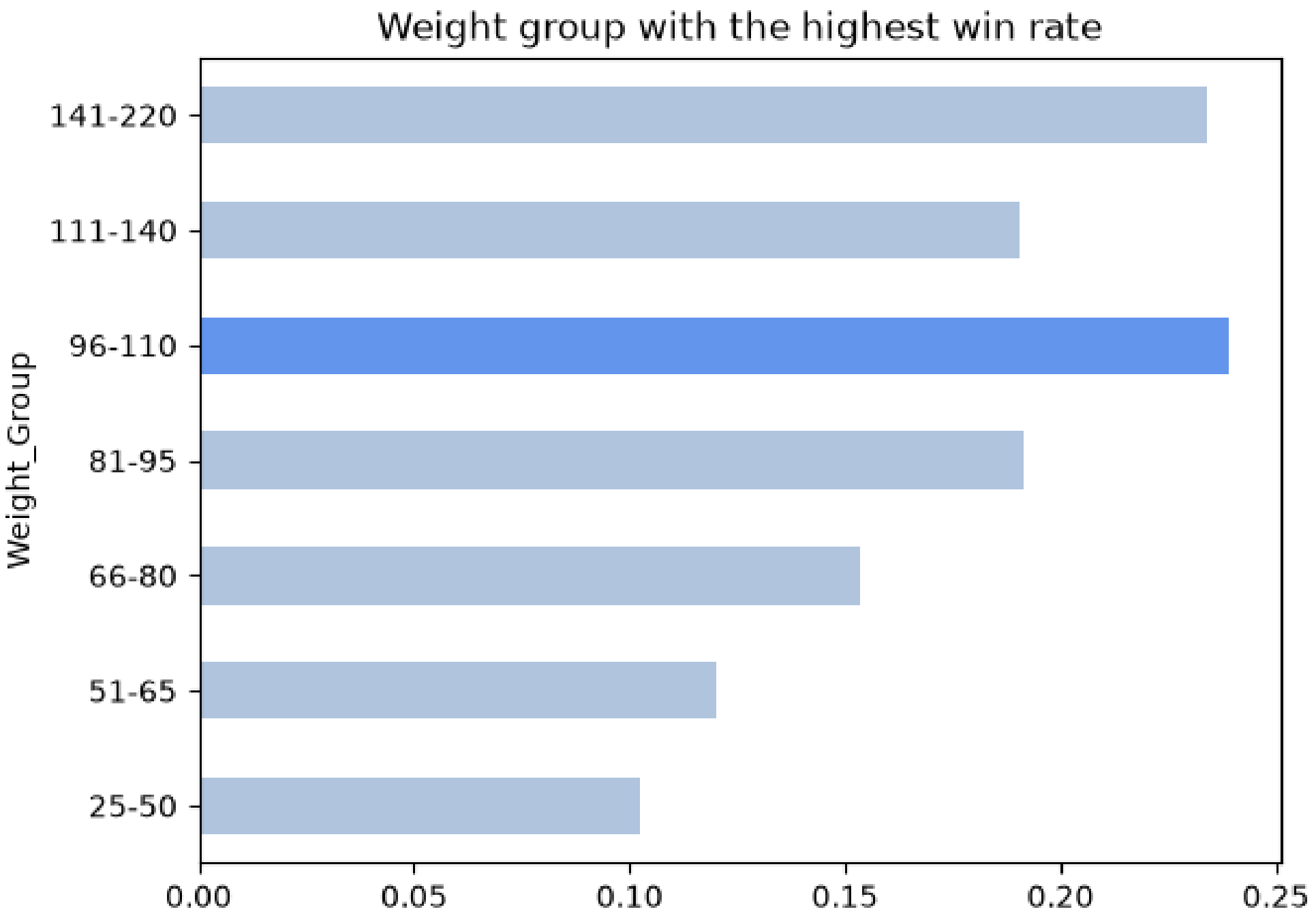
Winter Games win rate is about 11.8%.



It’s slightly easier to win at Summer.


Age group with the highest win rate
Age_Group
51+
41-50
31-40
26-30
18-25
Under 18
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
0.16

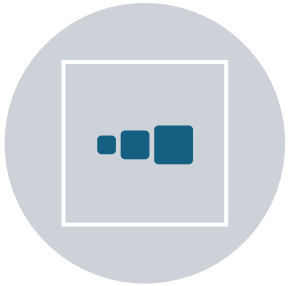

Height group with the highest win rate
Height_Group
211-230
196-210
181-195
166-180
151-165
125-150
0.00
0.05
0.10
0.15
0.20
0.25
0.30


Weight group with the highest win rate
Weight_Group
141-220
111-140
96-110
81-95
66-80
51-65
25-50
0.00
0.05
0.10
0.15
0.20
0.25

# Findings: body and age



**Age**: win rate peaks at 26-30 (about 17%) and is lowest under 18 and 51+ (about 10-11%).



**Height**: the tallest group (211-230 cm) wins about 32%, versus about 12% for the shortest.



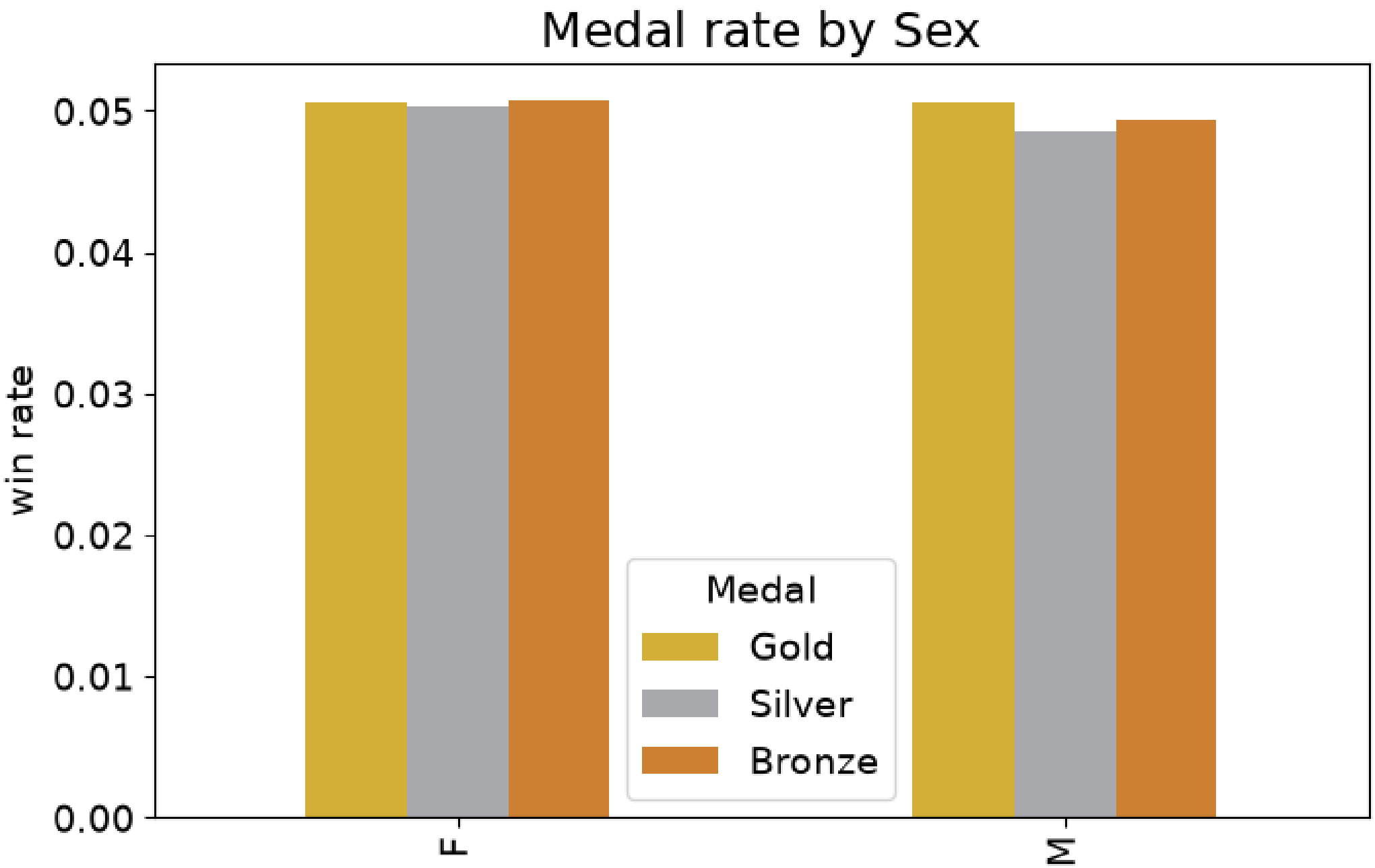
**Weight**: the 96-110 kg group peaks at about 24%.

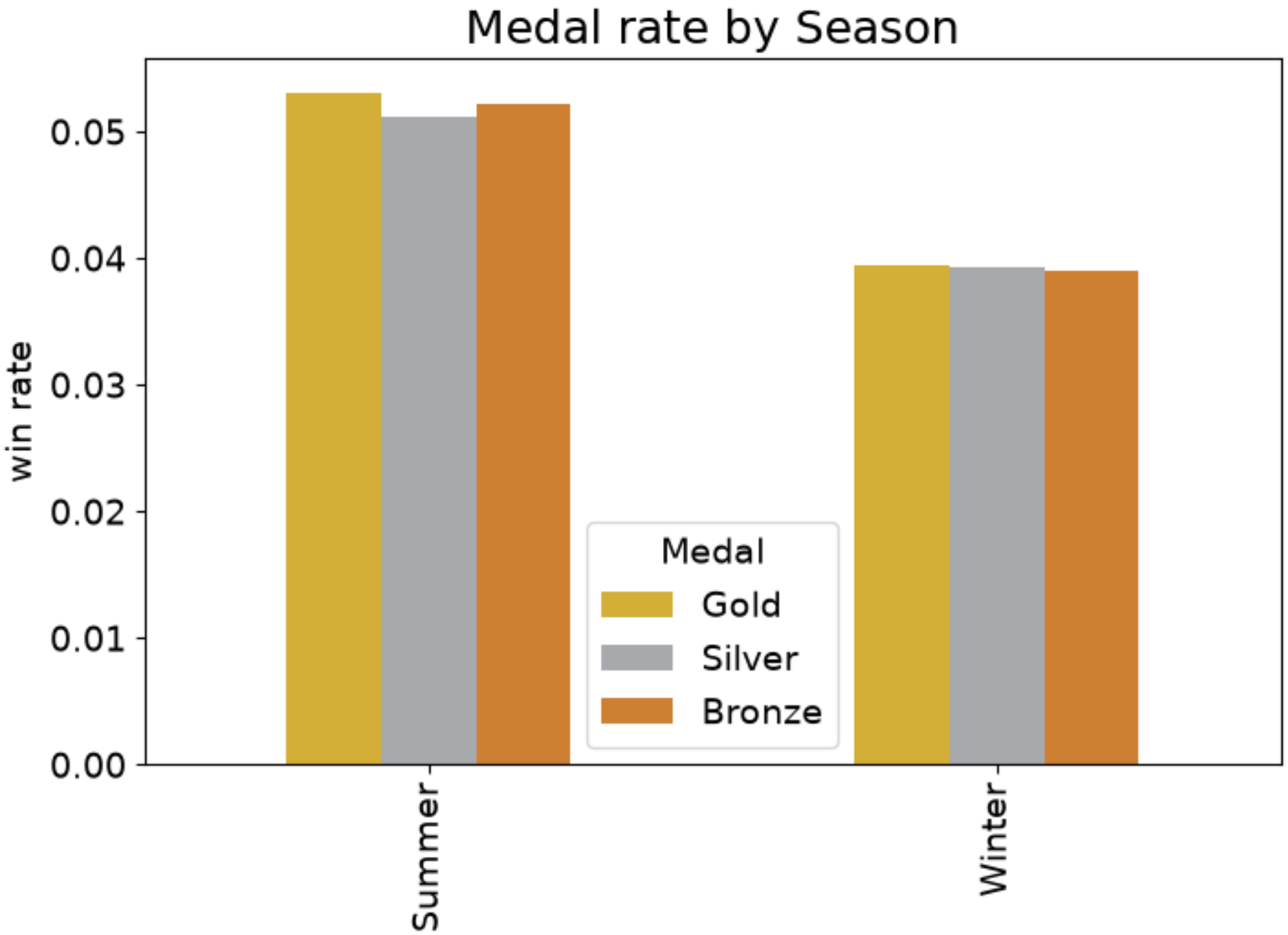


Tall and heavy athletes cluster in team sports, so this may reflect the sport, not the body.

Does each attribute change which medal an athlete wins?




Medal rate by Sex
win rate
0.05
0.04
0.03
0.02
0.01
0.00
F
M
Medal
Gold
Silver
Bronze

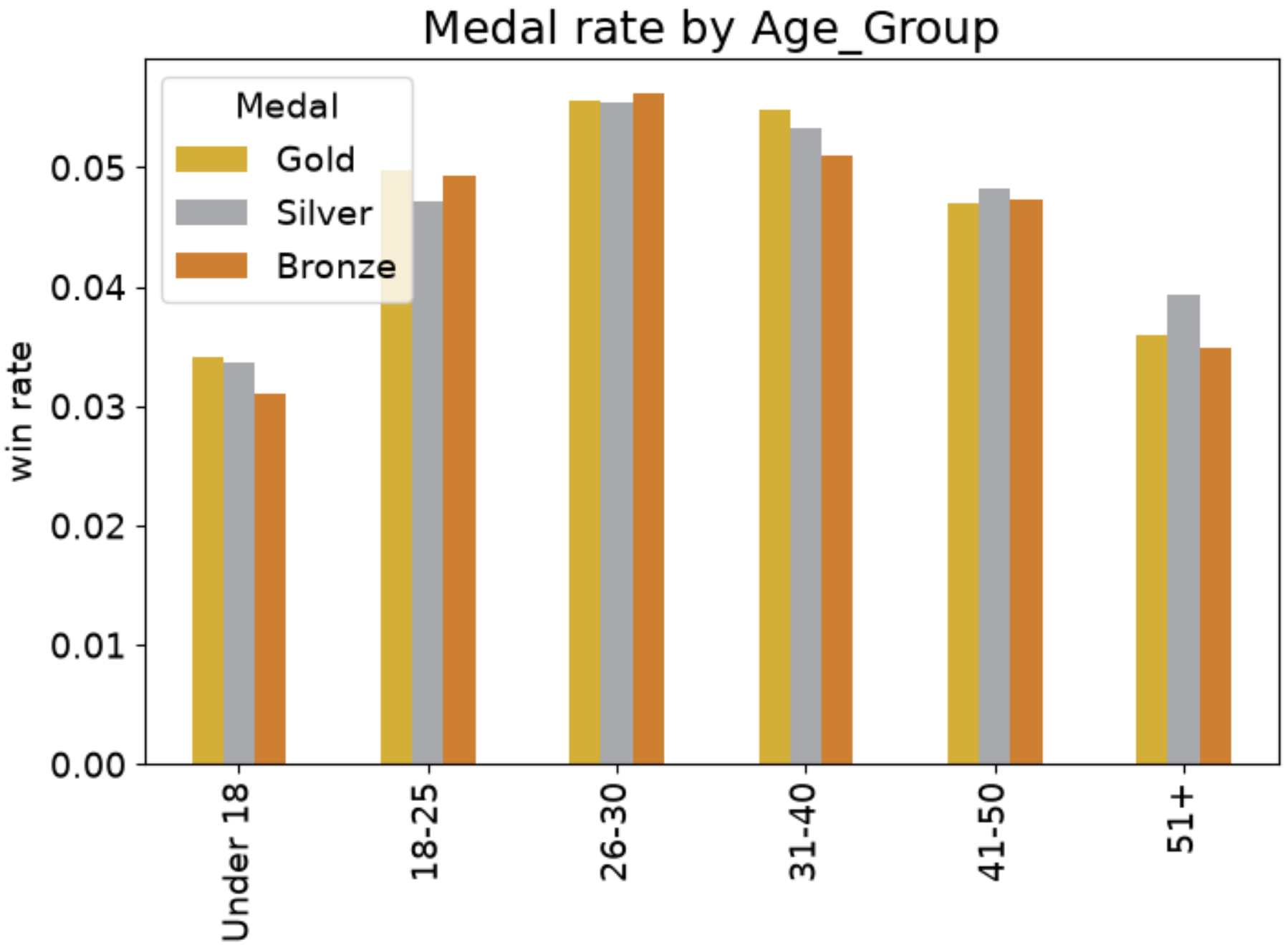

Medal rate by Season
win rate
0.05
0.04
0.03
0.02
0.01
0.00
Medal
Gold
Silver
Bronze
Summer
Winter

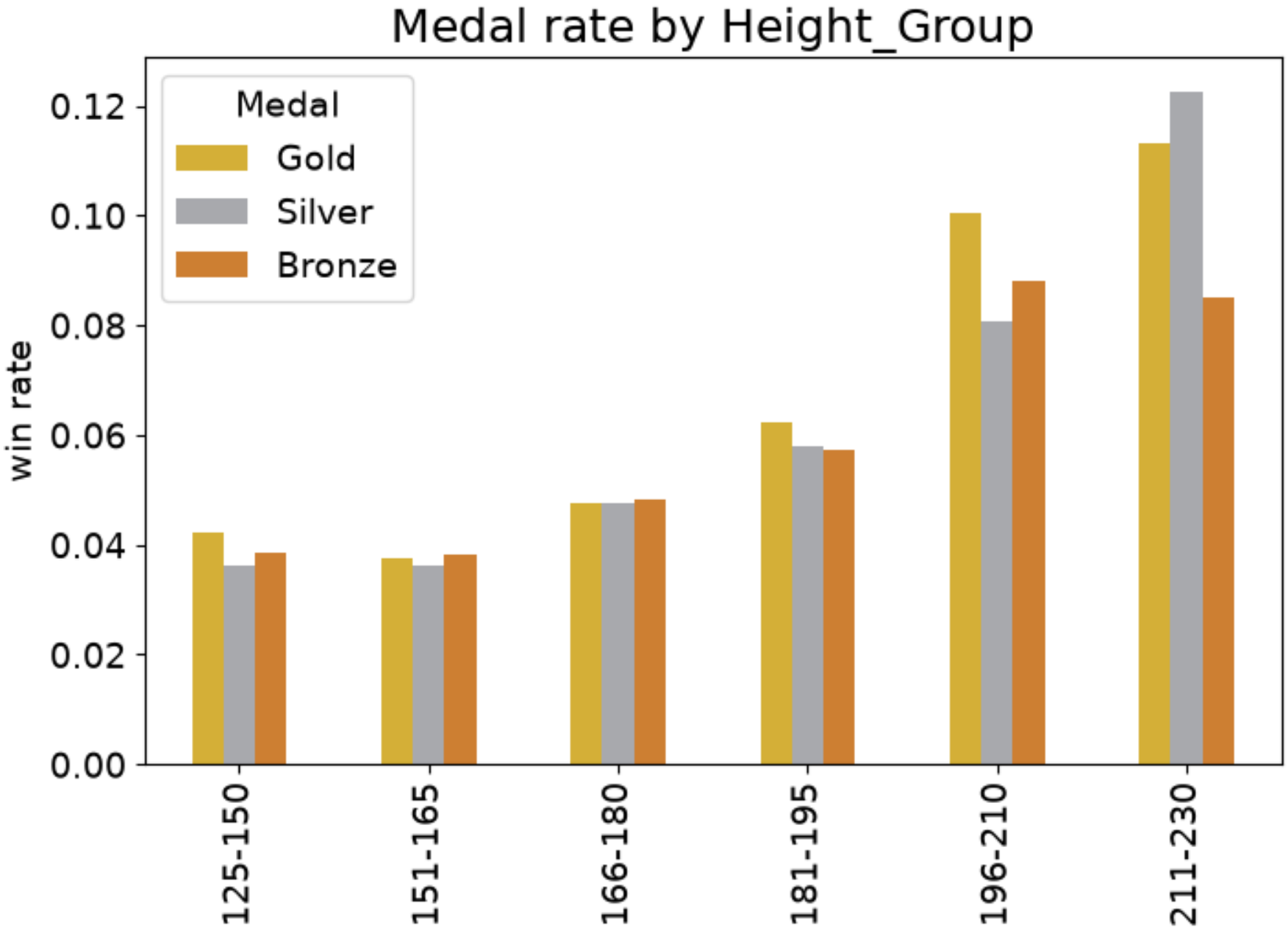
# Findings: sex and season

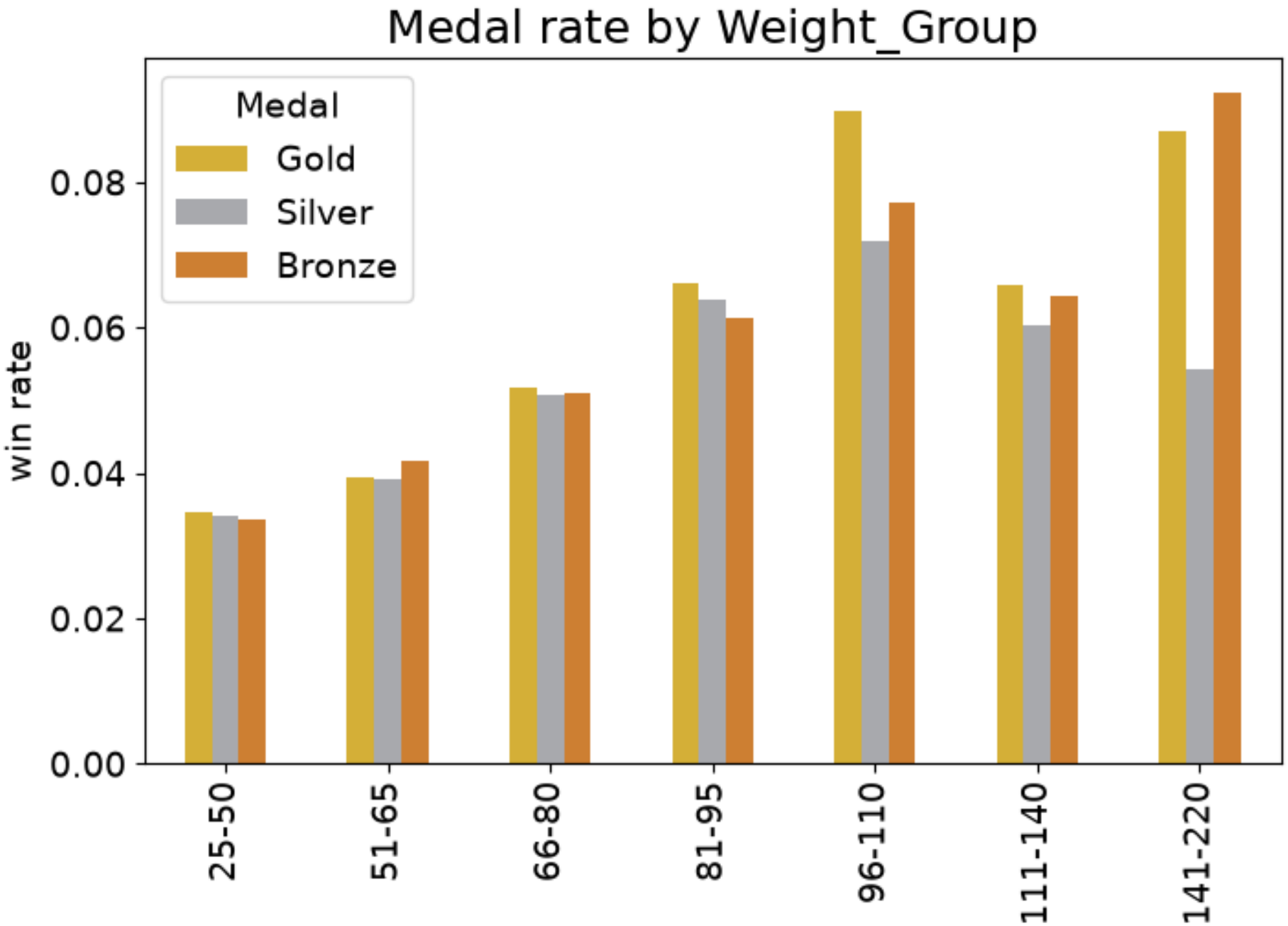
- **Sex:** Gold, Silver and Bronze are each about 5% for both women and men. Sex does not change the medal type.

- **Season:** Summer is about 5% for each medal and Winter about 4%, so Summer is higher overall.

- Within each season the three medals are **almost equal**, so season **changes how often**, not which medal.


Medal rate by Age_Group
Medal
Gold
Silver
Bronze
win rate
0.00
0.01
0.02
0.03
0.04
0.05
Under 18
18-25
26-30
31-40
41-50
51+


Medal rate by Height_Group
Medal
Gold
Silver
Bronze
win rate
0.12
0.10
0.08
0.06
0.04
0.02
0.00
125-150
151-165
166-180
181-195
196-210
211-230


Medal rate by Weight_Group
Medal
Gold
Silver
Bronze
win rate
0.00
0.02
0.04
0.06
0.08
25-50
51-65
66-80
81-95
96-110
111-140
141-220

# Findings: age, height and weight

- Age: all three medals peak at 26-30 (about 5.5%) and are lowest under 18 and 51+ (about 3-4%). The three medals stay close within each group.

- Height: medal rates climb with height, from about 4% (125-150 cm) to 9-12% (211-230 cm).

- Height and weight: the medal split becomes uneven only in the tallest and heaviest groups (for example Silver leads at 211-230 cm).

- Check group sizes and sport mix before trusting these gaps, since team sports may drive them.

# Conclusion

- Medal odds depend heavily on the sport, with a nearly 6x gap between the top and bottom.

- Team sports lead partly because one medal is shared across a roster.



# Thank You